



Chapter7 LTI Discrete-Time Systems in the Transform Domain

- Transfer Function Classification
- Types of Linear-Phase Transfer Functions
- Simple Digital Filters



Types of Transfer Functions



- The time-domain classification of a digital transfer function based on the length of its impulse response sequence:
 - Finite impulse response (FIR) transfer function.
 - Infinite impulse response (IIR) transfer function.

MOOC

Types of Transfer Functions



- In the case of digital transfer functions with frequency-selective frequency responses, there are two types of classifications:
 - (1) Classification based on the shape of the magnitude function $|H(e^{j\omega})|$.
 - (2) Classification based on the form of the phase function $\theta(\omega)$.

7.1 Transfer Function Classification Based on Magnitude Characteristics



- **Digital Filters with Ideal Magnitude Responses**
- **Bounded Real Transfer Function**
- **Allpass Transfer Function**

7.1.1 Digital Filters with Ideal Magnitude Responses



- A digital filter designed to pass signal components of certain frequencies without distortion should have a frequency response equal to 1 at these frequencies, and should have a frequency response equal to 0 at all other frequencies.

7.1.1 Digital Filters with Ideal Magnitude Responses

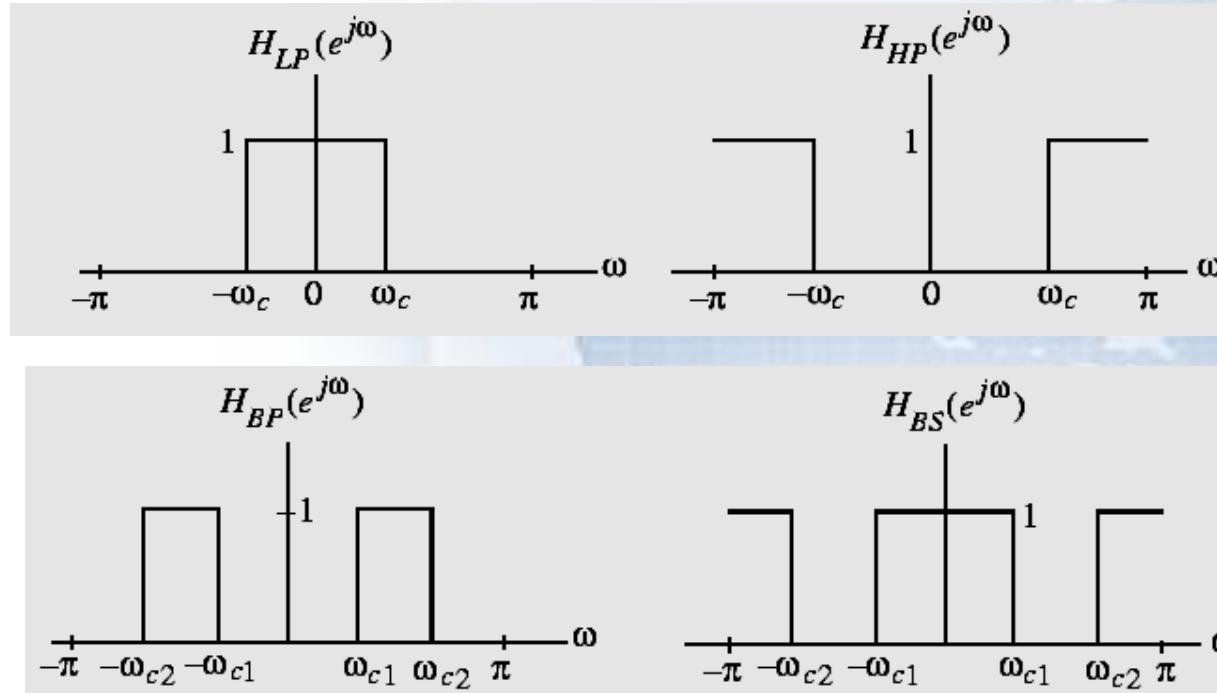


- The range of frequencies where the frequency response takes the value of 1 is called the **passband**.
- The range of frequencies where the frequency response takes the value of 0 is called the **stopband**.

7.1.1 Digital Filters with Ideal Magnitude Responses



- Magnitude responses of the four popular types of ideal digital filters with real impulse response coefficients are shown below:



7.1.1 Digital Filters with Ideal Magnitude Responses



- The frequencies ω_c , ω_{c1} , and ω_{c2} are called the **cutoff frequencies**.
- An ideal filter has a magnitude response equal to 1 in the passband and 0 in the stopband, and has a 0 phase everywhere.

7.1.1 Digital Filters with Ideal Magnitude Responses



- Earlier in the course we derived the inverse DTFT of the frequency response $H_{LP}(e^{j\omega})$ of the ideal lowpass filter:

$$h_{LP}[n] = \sin(\omega_c n) / (\pi n), \quad -\infty < n < \infty$$

- We have also shown that the above impulse response is not absolutely summable, and hence, the corresponding transfer function is not BIBO stable.
- Also, $h_{LP}[n]$ is not causal and is of doubly infinite length.

Ideal LPF is not stable and realizable!!

7.1.1 Digital Filters with Ideal Magnitude Responses



- The remaining three ideal filters are also characterized by doubly infinite, noncausal impulse responses and are not absolutely summable.
- Thus, the ideal filters with the ideal “**brick wall**” frequency responses cannot be realized with finite dimensional LTI filter.

7.1.1 Digital Filters with Ideal Magnitude Responses



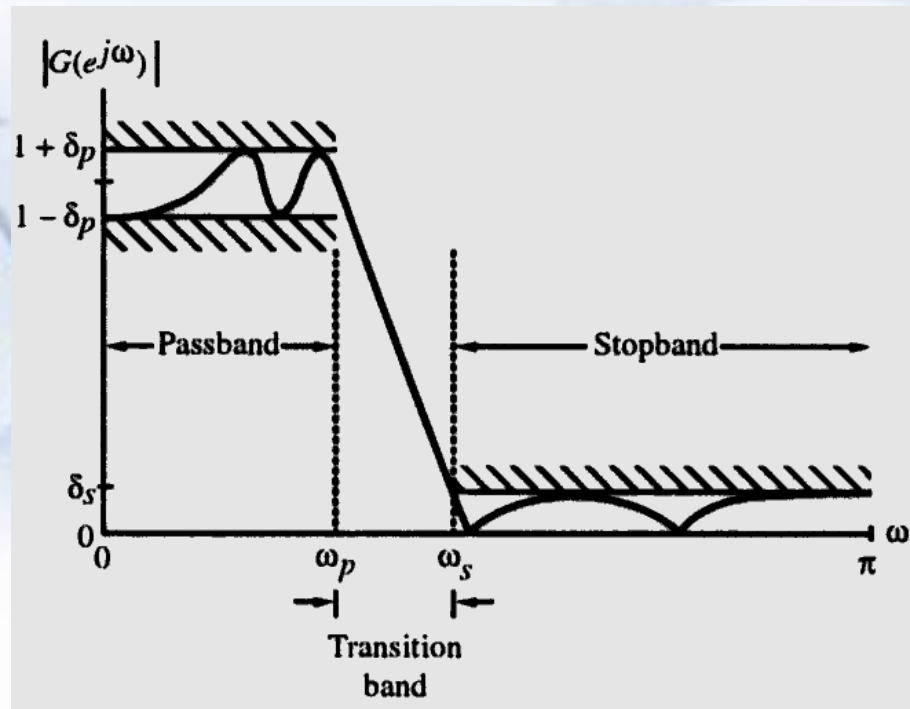
- To develop stable and realizable transfer functions, the ideal frequency response specifications are relaxed by including a **transition band** between the passband and the stopband.
- This permits the magnitude response to decay slowly from its maximum value in the passband to the 0 value in the stopband.

7.1.1 Digital Filters with Ideal Magnitude Responses



- Moreover, the magnitude response is allowed to vary by a small amount both in the passband and the stopband.

Typical magnitude response specifications of a lowpass filter:



7.1 Transfer Function Classification Based on Magnitude Characteristics



- Digital Filters with Ideal Magnitude Responses
- Bounded Real Transfer Function
- Allpass Transfer Function

7.1.2 Bounded Real Transfer Functions



- A causal stable real-coefficient transfer function $H(z)$ is defined as a **bounded real (BR)** transfer function if:

$$|H(e^{j\omega})| \leq 1 \quad \text{for all values of } \omega$$

- Let $x[n]$ and $y[n]$ denote, respectively, the input and output of a digital filter characterized by a BR transfer function $H(z)$ with $X(e^{j\omega})$ and $Y(e^{j\omega})$ denoting their DTFTs.

7.1.2 Bounded Real Transfer Functions



• Then the condition $|H(e^{j\omega})| \leq 1$ implies that:

$$|Y(e^{j\omega})|^2 \leq |X(e^{j\omega})|^2$$

Integrating the above from $-\pi$ to π , and applying Parseval's relation we get:

$$\sum_{n=-\infty}^{\infty} |y[n]|^2 \leq \sum_{n=-\infty}^{\infty} |x[n]|^2$$

7.1.2 Bounded Real Transfer Functions



- Thus, for all finite-energy inputs, the output energy is less than or equal to the input energy.
- It implies that a digital filter characterized by a BR transfer function can be viewed as a passive structure.

$|H(e^{j\omega})| = 1$, then the output energy is equal to the input energy, and such a digital filter is therefore a **lossless system**

7.1.2 Bounded Real Transfer Functions



- A causal stable real-coefficient transfer function $H(z)$ with $|H(e^{j\omega})|=1$ is thus called a **lossless bounded real (LBR) transfer function**
- The BR and LBR transfer functions are the keys to the realization of digital filters with low coefficient sensitivity.

7.1.2 Bounded Real Transfer Functions



- Example: Consider the causal stable IIR transfer function:

$$H(z) = \frac{K}{1 - \alpha z^{-1}}, \quad 0 < |\alpha| < 1$$

where K is a real constant.

- Its square-magnitude function is given by:

$$|H(e^{j\omega})|^2 = H(z)H(z^{-1})\big|_{z=e^{j\omega}} = \frac{K^2}{(1 + \alpha^2) - 2\alpha \cos \omega}$$

7.1.2 Bounded Real Transfer Functions



- Thus, for $\alpha > 0$,

$$(|H(e^{j\omega})|^2)_{\max} = K^2/(1-\alpha)^2 \mid_{\omega=0}$$

$$(|H(e^{j\omega})|^2)_{\min} = K^2/(1+\alpha)^2 \mid_{\omega=\pi}$$

- On the other hand, for $\alpha < 0$,

$$(|H(e^{j\omega})|^2)_{\max} = K^2/(1+\alpha)^2 \mid_{\omega=\pi}$$

$$(|H(e^{j\omega})|^2)_{\min} = K^2/(1-\alpha)^2 \mid_{\omega=0}$$

7.1.2 Bounded Real Transfer Functions



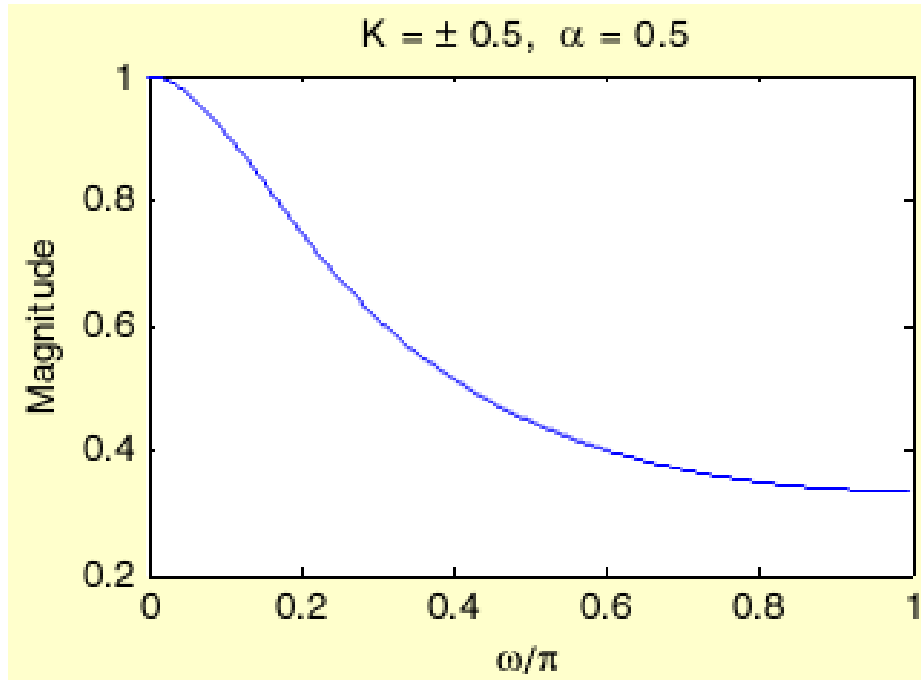
- Hence,

$$H(z) = \frac{K}{1 - \alpha z^{-1}}, \quad 0 < |\alpha| < 1$$

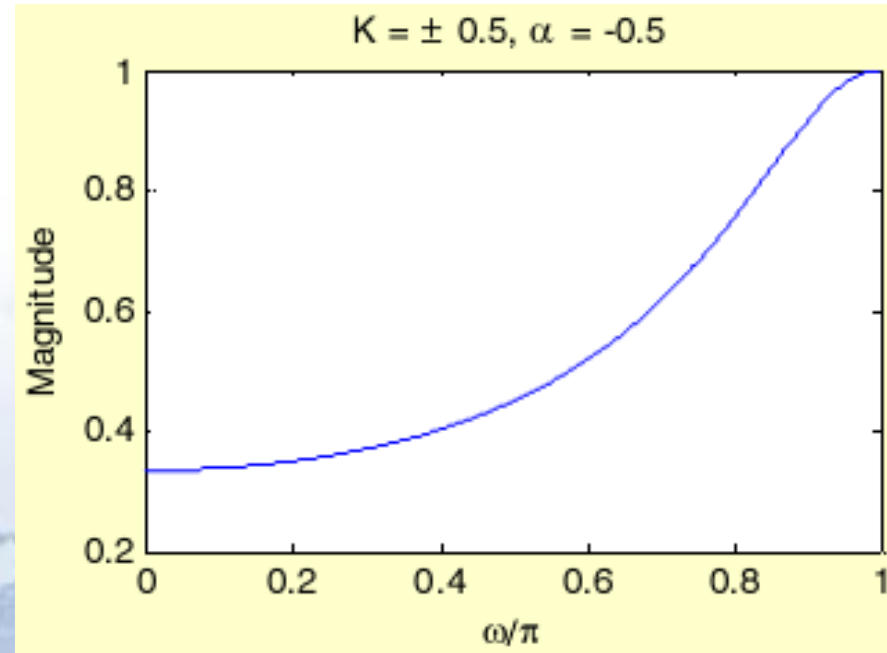
is a BR function for $K = \pm(1 - \alpha)$,

Plots of the magnitude function for $\alpha = \pm 0.5$ with values of K chosen to make $H(z)$ a BR function are shown on the next page.

7.1.2 Bounded Real Transfer Functions



Lowpass filter



Highpass filter

7.1 Transfer Function Classification Based on Magnitude Characteristics



- Digital Filters with Ideal Magnitude Responses
- Bounded Real Transfer Function
- Allpass Transfer Function

7.1.3 Allpass Transfer Function (difficulty)



- The magnitude response of *allpass system* satisfies:

$$|A(e^{j\omega})|^2=1, \text{ for all } \omega.$$

- Example:

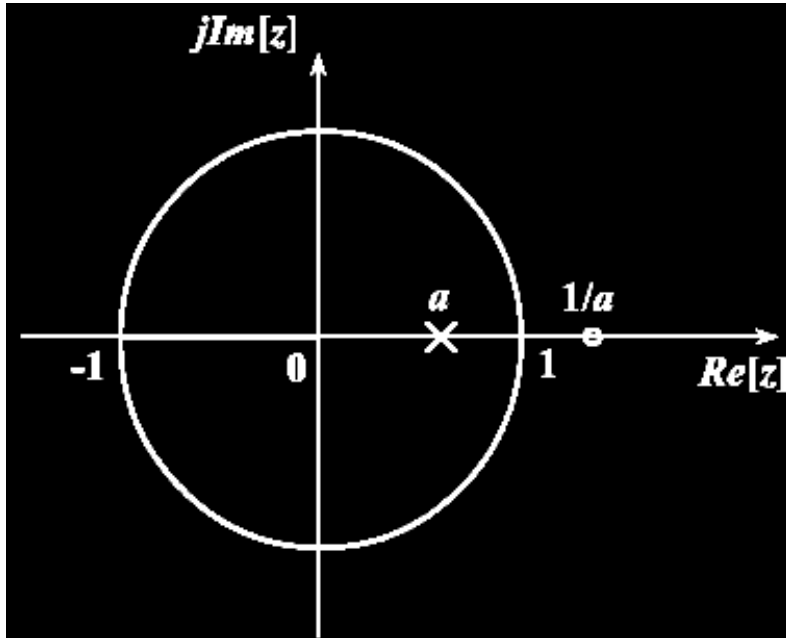
– The $A(z)$ of a simple 1st-order allpass system is:

$$A(z) = \frac{z^{-1} - a}{1 - az^{-1}}$$

where a is real, and $0 < |a| < 1$. When a is complex, the $A(z)$ should be:

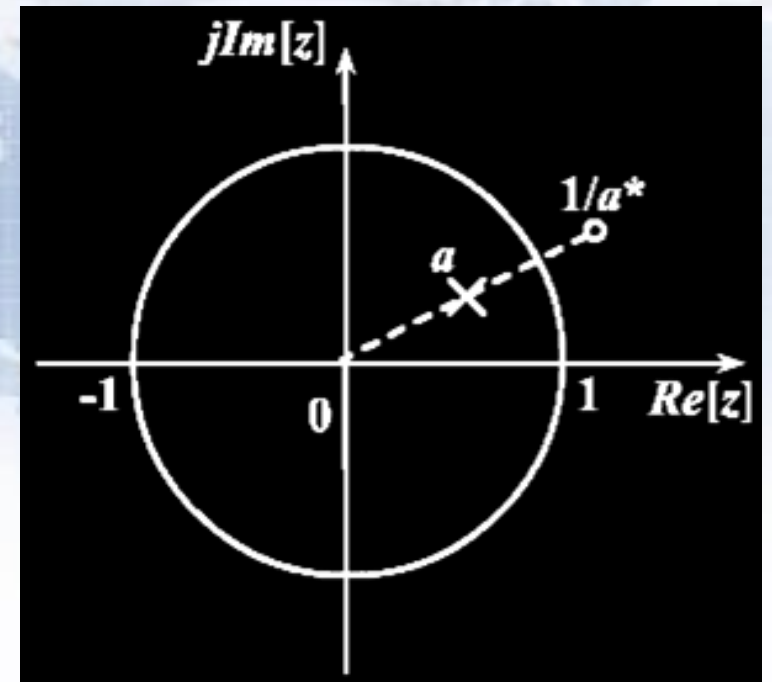
$$A(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$$

7.1.3 Allpass Transfer Function



one real pole

one complex pole



7.1.3 Allpass Transfer Function

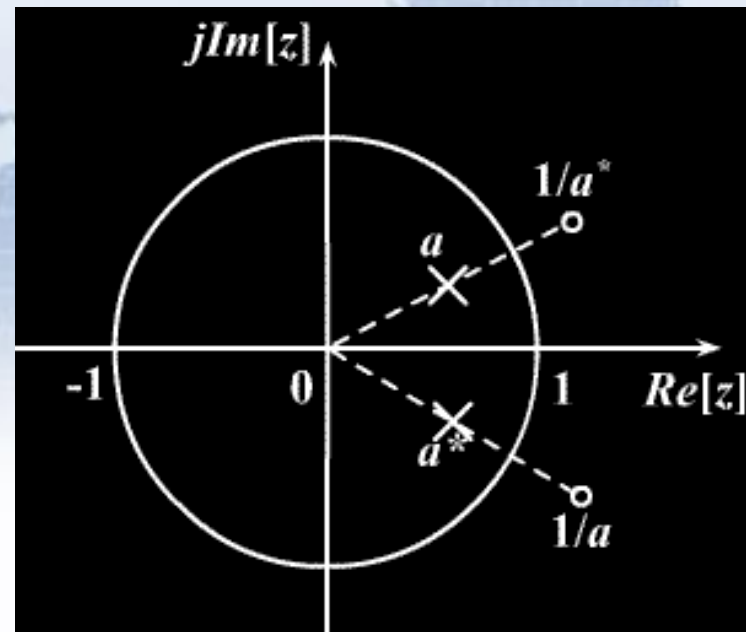


- 2nd-order allpass transfer function

$$H_{ap}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}} \cdot \frac{z^{-1} - a}{1 - a^* z^{-1}}$$

poles: $z = a, a^*$

zeros: $z = 1/a^*, 1/a$



7.1.3 Allpass Transfer Function



- The M th-order causal real-coefficient allpass system is

$$A_M(z) = \pm \frac{d_M + d_{M-1}z^{-1} + \dots + d_1z^{-M+1} + z^{-M}}{1 + d_1z^{-M} + \dots + d_{M-1}z^{-M+1} + d_Mz^{-M}}$$

- It can also be expressed in a factored form

$$A_M(z) = \pm \prod_{i=1}^M \left(\frac{-\lambda_i^* + z^{-1}}{1 - \lambda_i z^{-1}} \right)$$

- If we denote polynomial: $D_M(z) = 1 + d_1z^{-1} + \dots + d_{M-1}z^{-M+1} + d_Mz^{-M}$

$$A_M(z) = \pm \frac{z^{-M} D_M(z^{-1})}{D_M(z)}$$

7.1.3 Allpass Transfer Function



- The numerator of a real-coefficient allpass transfer function is said to be the mirror-image polynomial of the denominator, and vice versa.
- We shall use the notation $\tilde{D}_M(z)$ to denote the **mirror-image polynomial** of a degree- M polynomial $D_M(z)$, i.e.,

$$\tilde{D}_M(z) = z^{-M} D_M(z^{-1})$$

7.1.3 Allpass Transfer Function

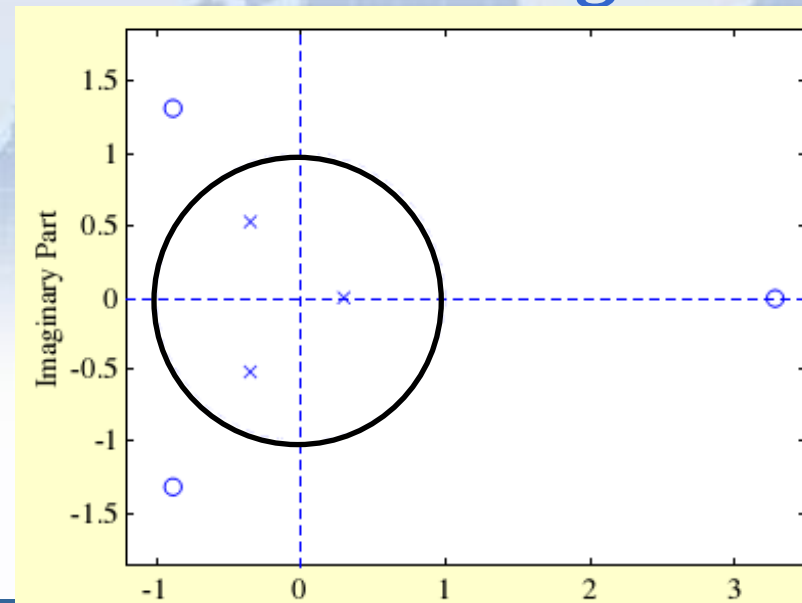


- The expression

$$A_M(z) = \pm \frac{z^{-M} D_M(z^{-1})}{D_M(z)}$$

implies that the poles and zeros of a real-coefficient allpass function exhibit mirror-image symmetry in the z -plane.

$$A_3(z) = \frac{-0.2 + 0.18z^{-1} + 0.4z^{-2} + z^{-3}}{1 + 0.4z^{-1} + 0.18z^{-2} - 0.2z^{-3}}$$



7.1.3 Allpass Transfer Function



- To show that $|A_M(e^{j\omega})|=1$ we observe that:

$$A_M(z^{-1}) = \pm \frac{z^M D_M(z)}{D_M(z^{-1})}$$

- Therefore:

$$A_M(z)A_M(z^{-1}) = \frac{z^{-M} D_M(z^{-1})}{D_M(z)} \frac{z^M D_M(z)}{D_M(z^{-1})}$$

Hence:

$$|A_M(e^{j\omega})|^2 = A_M(z)A_M(z^{-1}) \Big|_{z=e^{j\omega}} = 1$$

7.1.3 Allpass Transfer Function



- Properties:
 - (1) A causal stable real-coefficient allpass transfer function is a **lossless** bounded real (LBR) function or, equivalently, a causal stable allpass filter is a lossless structure.
 - (2) The magnitude function of a stable allpass function $A(z)$ satisfies:

$$|A(z)| \begin{cases} < 1, & \text{for } |z| > 1 \\ = 1, & \text{for } |z| = 1 \\ > 1, & \text{for } |z| < 1 \end{cases}$$

7.1.3 Allpass Transfer Function



(3) Let $\tau_g(\omega)$ denote the group delay function of an allpass filter $A(z)$, i.e.,

$$\tau_g(\omega) = -\frac{d}{d\omega}[\theta_c(\omega)]$$

The unwrapped phase function $\theta_c(\omega)$ of a stable allpass function is a ***monotonically decreasing function*** of ω so that $\tau_g(\omega)$ is everywhere ***positive*** in the range $0 < \omega < \pi$.

Application of allpass system



- Any causal stable system can be denoted as:

$$H(z)=H_{\min}(z)A(z)$$

where $H_{\min}(z)$ is a minimum phase-delay system.

- Use allpass system to help design stable filters.
- Use allpass system to help design linear phase system.
- A simple example.(Fig7.7)

Application of allpass system

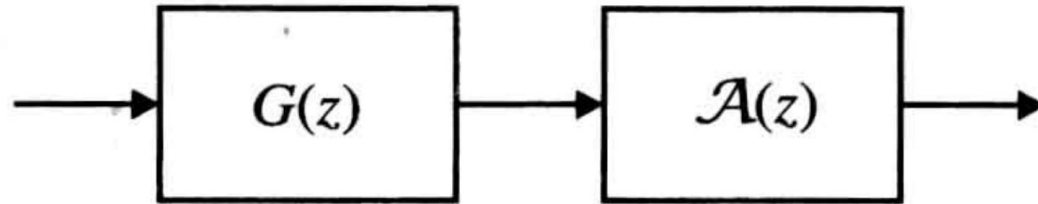


Figure 7.7: Use of an allpass filter as a delay equalizer.

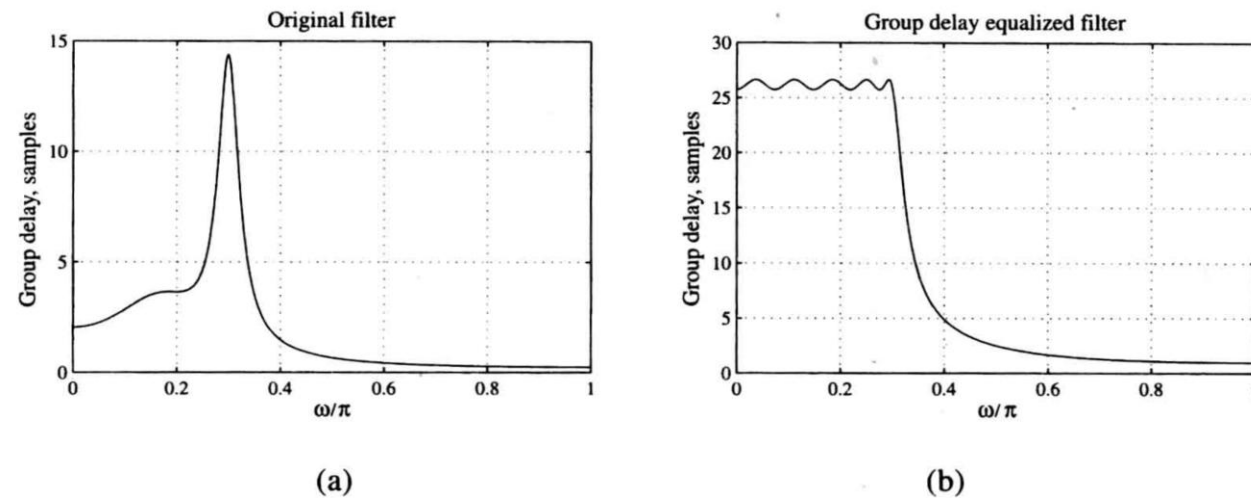


Figure 7.8: (a) The group delay of the elliptic filter and (b) the group delay of the cascade of the original filter and the allpass section.

Types of Transfer Functions

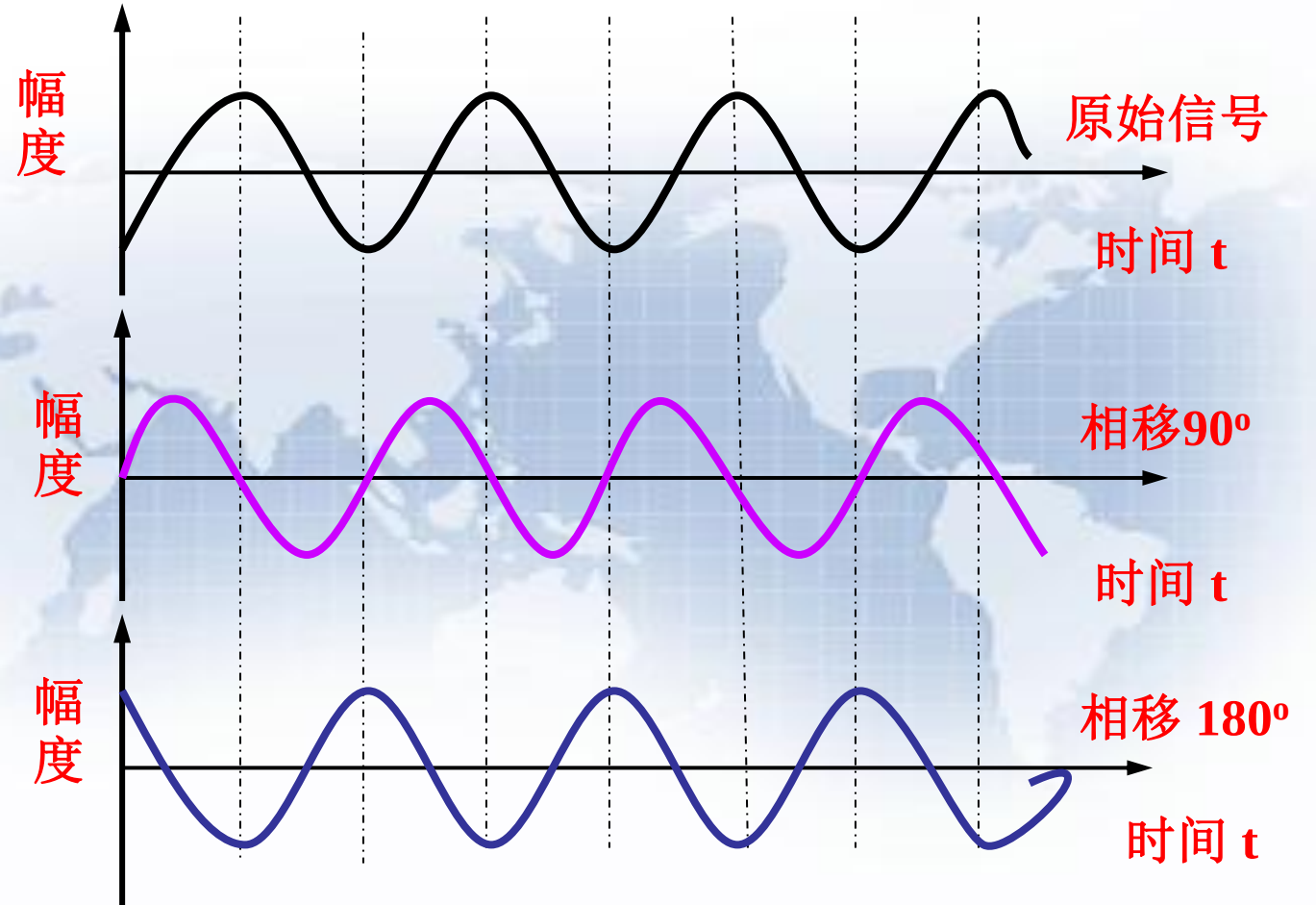


- In the case of digital transfer functions with frequency-selective frequency responses, there are two types of classifications:
 - (1) Classification based on the shape of the magnitude function $|H(e^{i\omega})|$.
 - (2) Classification based on the form of the phase function $\theta(\omega)$.

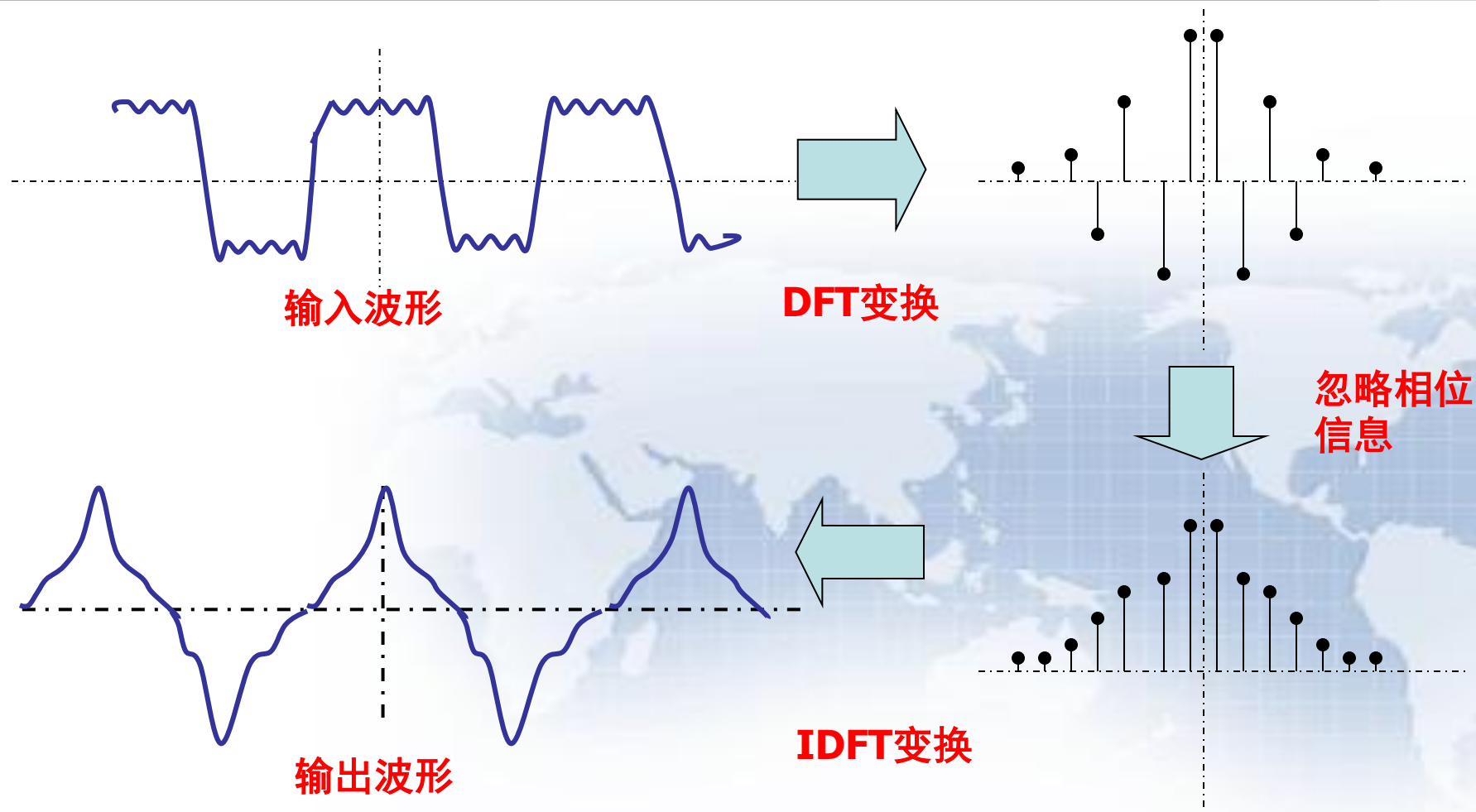
7.2 Transfer Function Classification Based on Phase Characteristic



1、The phase delay will cause the change of signal waveform



2、 If we ignore the phase information, then

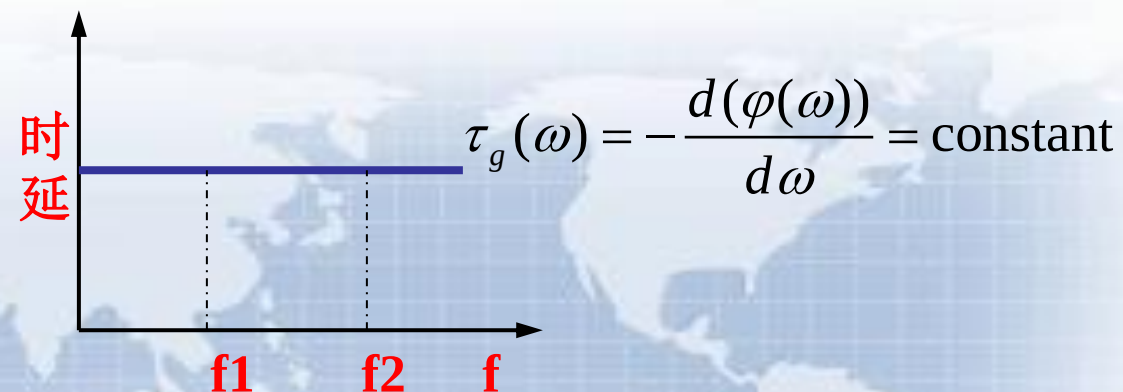
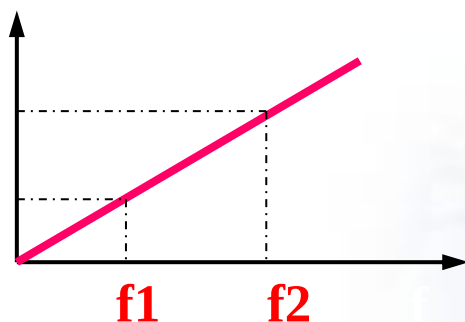


3、 The nonlinearity of system phase delay will cause the signal distortion

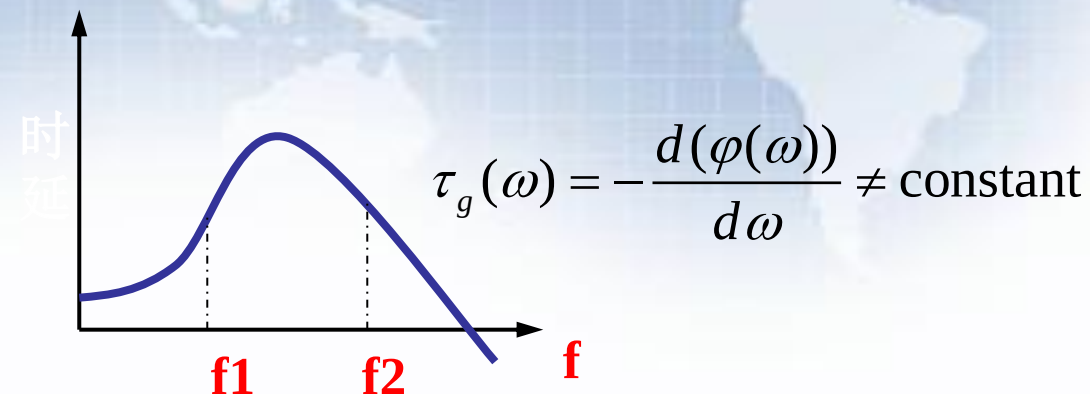
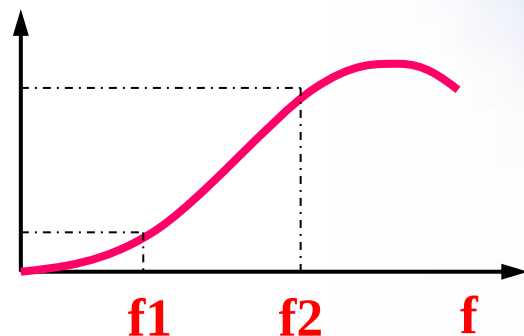


Time delay of signal depends on system phase characteristic

- linear phase



- Nonlinear phase



7.2 Transfer Function Classification Based on Phase Characteristic



- **Zero-Phase Transfer Function**
- **Linear-Phase Transfer Function**
- **Minimum-Phase and Maximum-Phase Transfer Functions**

7.2.1 Zero-Phase Transfer Function



- One way to avoid any phase distortions is to make the frequency response of the filter real and nonnegative, to design the filter with a **zero phase characteristic**.

$$H(e^{j\omega}) = |H(e^{j\omega})|$$

- But for a causal digital filter it is **impossible**.

7.2.1 Zero-Phase Transfer Function

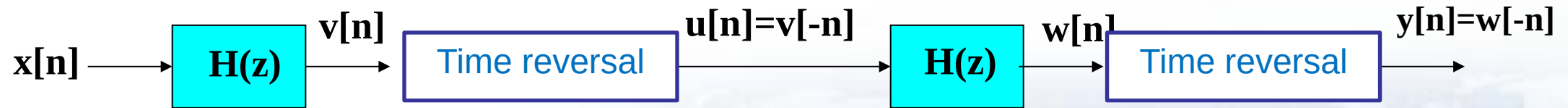


- Only for non-real-time processing of real-valued input signals of finite length, the zero phase condition can be met.
- Let $H(z)$ be a **real-coefficient** rational z -transform with no poles on the unit cycle, then $F(z)=H(z)H(z^{-1})$ has a zero phase on the unit cycle.

7.2.1 Zero-Phase Transfer Function



- A simple realization of zero-phase transfer function



➤ **$h[n]$: Causal and real-coefficient**

$$V(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$U(e^{j\omega}) = V^*(e^{j\omega}) = H^*(e^{j\omega})X^*(e^{j\omega})$$

$$W(e^{j\omega}) = H(e^{j\omega})U(e^{j\omega}) = |H(e^{j\omega})|^2 X^*(e^{j\omega})$$

$$Y(e^{j\omega}) = W^*(e^{j\omega}) = |H(e^{j\omega})|^2 X(e^{j\omega})$$

- The function **filtfilt** implements the above zero-phase filtering scheme.

7.2.2 Linear-Phase Transfer Function **key point**



- The phase distortion can be avoided by ensuring that the transfer function has a unity magnitude and a linear-phase characteristic, that is:

$$H(e^{j\omega}) = e^{-j\omega D}$$

- How to perform the linear-phase filter?

$$y[n] = x[n-D]$$

$$\rightarrow Y(e^{j\omega}) = e^{-j\omega D} X(e^{j\omega})$$

$$\rightarrow H(e^{j\omega}) = Y(e^{j\omega}) / X(e^{j\omega}) = e^{-j\omega D}$$

7.2.2 Linear-Phase Transfer Function **key point**



- Example - Determine the impulse response of an ideal lowpass filter with a linear phase response:

$$H_{LP}(e^{j\omega}) = \begin{cases} e^{-j\omega n_o}, & 0 < |\omega| < \omega_c \\ 0, & \omega_c \leq |\omega| \leq \pi \end{cases}$$

- Applying the frequency-shifting property of the DTFT to the impulse response of an ideal zero-phase lowpass filter we arrive at:

$$h_{LP}[n] = \frac{\sin \omega_c (n - n_o)}{\pi (n - n_o)}, \quad -\infty < n < \infty$$

7.2.2 Linear-Phase Transfer Function **key point**



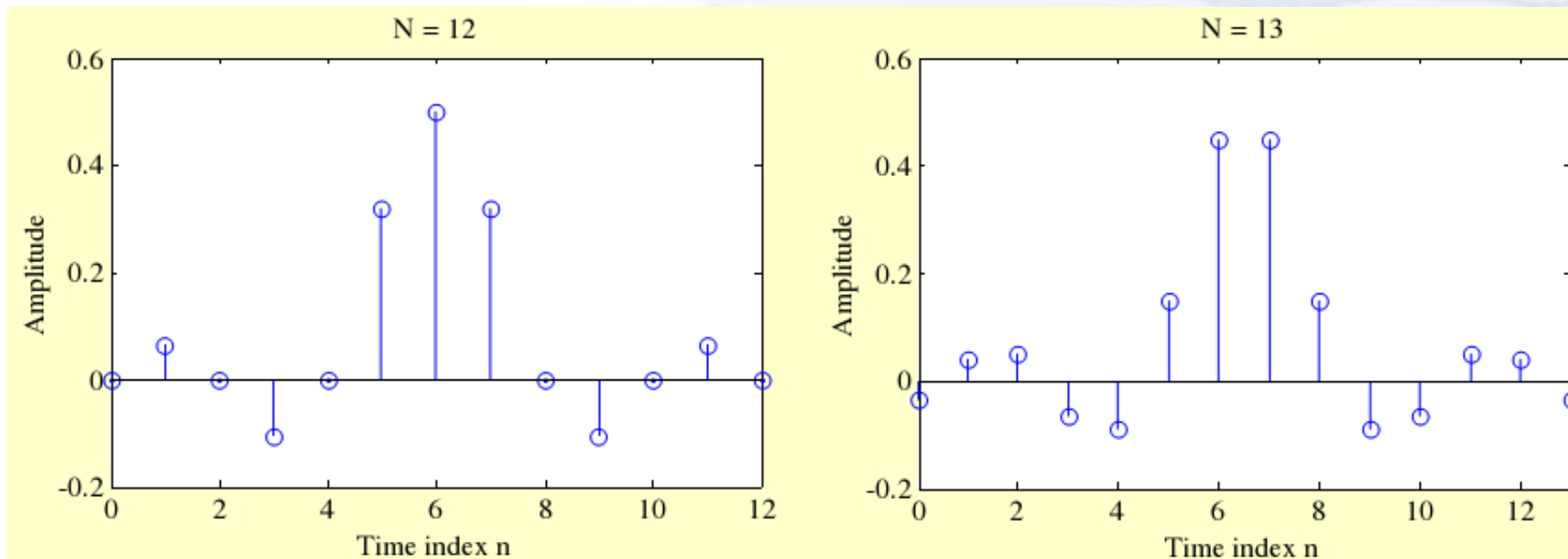
- By truncating the impulse response to a finite number of terms, a realizable FIR approximation to the ideal lowpass filter can be developed.
- The truncated approximation may or may not exhibit linear phase, depending on the value of n_0 chosen.
- If we choose $n_0 = N/2$ with N a positive integer, the truncated and shifted approximation:

$$h_{LP}[n] = \frac{\sin \omega_c (n - N/2)}{\pi (n - N/2)}, \quad 0 \leq n \leq N$$

7.2.2 Linear-Phase Transfer Function **key point**



- Figure below shows the filter coefficients obtained using the function **sinc** for two different values of N .



7.2.3 Minimum-Phase and Maximum-Phase Transfer Function (difficulty)



- Based on the expression:

$$H(e^{j\omega}) = \frac{p_0}{d_0} e^{j\omega(N-M)} \frac{\prod_{k=1}^M (e^{j\omega} - \xi_k)}{\prod_{k=1}^N (e^{j\omega} - \lambda_k)} = |H(e^{j\omega})| e^{j \arg[H(e^{j\omega})]}$$

The phase function is:

$$\arg \frac{H(e^{j\omega})}{p_0 / d_0} = \omega(N-M) + \sum_{k=1}^M \arg(e^{j\omega} - \xi_k) - \sum_{k=1}^N \arg(e^{j\omega} - \lambda_k)$$

ξ_k and λ_k are zeros and poles, respectively.

7.2.3 Minimum-Phase and Maximum-Phase Transfer Function **difficulty**



- We define the expression $(e^{j\omega} - \xi_k)$ and $(e^{j\omega} - \lambda_k)$ as *zero vectors* and *pole vectors*.
- When ξ_k and λ_k are **inside** the unit circle, and ω change from 0 to 2π , the change of phase of the zero (pole) vectors are 2π .
- When ξ_k and λ_k are **outside** the unit circle, and ω change from 0 to 2π , the change of phase of the zero (pole) vectors are 0.

7.2.3 Minimum-Phase and Maximum-Phase Transfer Function **difficulty**



- So, only zeros or poles inside the unit circle can affect phase function of $H(e^{j\omega})$.
- For **causal stable** system, we can deduce:

$$\Delta \arg \frac{H(e^{j\omega})}{p_0 / d_0} \Big|_{\Delta\omega=2\pi} \leq 0$$

So, it's the **phase delay (lag)** system.



- When all **zeros** are all **inside** the unit circle, we get:

$$\Delta \arg \frac{H(e^{j\omega})}{p_0 / d_0} \Big|_{\Delta\omega=2\pi} = 0$$

It's the **minimum phase delay system**.

7.2.3 Minimum-Phase and Maximum-Phase Transfer Function **difficulty**



- When **all zeros** are all **outside** the unit circle, we get:

$$\Delta \arg \frac{H(e^{j\omega})}{p_0 / d_0} \Big|_{\Delta\omega=2\pi} = -2\pi M$$

where M is the number of zeros. It's the **maximum phase delay system**.

- A transfer function with zeros inside and outside the unit circle is called a **mixed-phase transfer function**.
- Example 7.4

7.2.3 Minimum-Phase and Maximum-Phase Transfer Function **difficulty**



- **Example 7.4**

$$H(z) = \frac{2(1 + 0.3z^{-1})(0.4 - z^{-1})}{(1 - 0.2z^{-1})(1 + 0.5z^{-1})}$$

- **It's mixed-phase transfer function. On the other hand, notice that**

$$H(z) = \underbrace{\left[\frac{2(1 + 0.3z^{-1})(1 - 0.4z^{-1})}{(1 - 0.2z^{-1})(1 + 0.5z^{-1})} \right]}_{\text{minimum-phase transfer function}} \underbrace{\left[\frac{(0.4 - z^{-1})}{(1 - 0.4z^{-1})} \right]}_{\text{Allpass transfer function}}$$

$$H(z) = \underbrace{\left[\frac{2(0.3 + z^{-1})(0.4 - z^{-1})}{(1 - 0.2z^{-1})(1 + 0.5z^{-1})} \right]}_{\text{maximum-phase transfer function}} \underbrace{\left[\frac{(1 + 0.3z^{-1})}{(0.3 + z^{-1})} \right]}_{\text{Allpass transfer function}}$$

7.3 Types of Linear-Phase FIR Transfer Functions



- It is nearly impossible to design a linear-phase IIR transfer function.
- It is always possible to design an FIR transfer function with an exact linear-phase response.

Under what conditions an FIR transfer function is linear-phase?



7.3 Types of Linear-Phase FIR Transfer Functions



- Consider a causal FIR transfer function $H(z)$ of length $N+1$, i.e., of order N :

$$H(z) = \sum_{n=0}^N h[n] z^{-n}$$

- If $H(z)$ is to have a linear-phase, its frequency response must be of the form

$$H(e^{j\omega}) = e^{j(c\omega + \beta)} \check{H}(\omega)$$

where c and β are constants, and $\check{H}(\omega)$, called the **amplitude response**, also called the **zero-phase response**, is a real function of ω .

7.3 Types of Linear-Phase FIR Transfer Functions



- For a **real** impulse response, the magnitude response $|H(e^{j\omega})|$ is an even function of ω , i.e.,

$$|H(e^{j\omega})| = |H(e^{-j\omega})|$$

- Since $|H(e^{j\omega})| = |\check{H}(\omega)|$ and $|H(e^{-j\omega})| = |\check{H}(-\omega)|$

the amplitude response is then either an even function or an odd function of ω , i.e.

$$\check{H}(-\omega) = \pm \check{H}(\omega)$$

7.3 Types of Linear-Phase FIR Transfer Functions



- The frequency response satisfies the relation $H(e^{j\omega})=H^*(e^{-j\omega})$, or equivalently, the relation

$$e^{j(c\omega+\beta)}\breve{H}(\omega)=e^{-j(-c\omega+\beta)}\breve{H}(-\omega)$$

- If $\breve{H}(\omega)$ is an **even** function, then the above relation leads to $e^{j\beta}=e^{-j\beta}$ implying that

$$\beta=0 \text{ or } \beta=\pi$$

7.3 Types of Linear-Phase FIR Transfer Functions



- From

$$H(e^{j\omega}) = e^{j(c\omega + \beta)} \check{H}(\omega)$$

$$\check{H}(\omega) = e^{-j(c\omega + \beta)} H(e^{j\omega})$$

Substituting the value of β in the above we get

$$\check{H}(\omega) = \pm e^{-jc\omega} H(e^{j\omega}) = \pm \sum_{n=0}^N h[n] e^{-j\omega(c+n)}$$

7.3 Types of Linear-Phase FIR Transfer Functions



- Replacing ω with $-\omega$ in the previous equation we get

$$\check{H}(-\omega) = \pm \sum_{l=0}^N h[l] e^{j\omega(c+l)}$$

Making a change of variable $l=N-n$, we rewrite the above equation as

$$\check{H}(-\omega) = \pm \sum_{n=0}^N h[N-n] e^{j\omega(c+N-n)}$$

7.3 Types of Linear-Phase FIR Transfer Functions



As $\check{H}(\omega) = \check{H}(-\omega)$, we have

$$h[n]e^{-j\omega(c+n)} = h[N-n]e^{j\omega(c+N-n)}$$

- The above leads to the condition

$$c = -N/2 \text{ and } h[n] = h[N-n], 0 \leq n \leq N$$

- Thus, the FIR filter with an even amplitude response will have a linear phase if it has a **symmetric** impulse response.

7.3 Types of Linear-Phase FIR Transfer Functions



If $\check{H}(\omega)$ is an **odd** function of ω , then from

$$e^{j(c\omega+\beta)}\check{H}(\omega)=e^{-j(-c\omega+\beta)}\check{H}(-\omega)$$

We get $e^{j\beta} = -e^{-j\beta}$ as $\check{H}(-\omega) = -\check{H}(\omega)$

The above is satisfied if $\beta = \pi/2$ or $\beta = -\pi/2$, Then

$$H(e^{j\omega}) = e^{j(c\omega+\beta)}\check{H}(\omega)$$

$$H(e^{j\omega}) = je^{jc\omega}\check{H}(\omega)$$

7.3 Types of Linear-Phase FIR Transfer Functions



- The last equation can be rewritten as:

$$\check{H}(\omega) = -je^{-jc\omega} H(e^{j\omega}) = -j \sum_{n=0}^N h[n] e^{-j\omega(c+n)}$$

As $\check{H}(-\omega) = -\check{H}(\omega)$, from the above we get

$$\check{H}(-\omega) = j \sum_{l=0}^N h[l] e^{j\omega(c+l)}$$

7.3 Types of Linear-Phase FIR Transfer Functions



- Making a change of variable $l=N-n$, we rewrite the last equation as:

$$\tilde{H}(-\omega) = j \sum_{n=0}^N h[N-n] e^{j\omega(c+N-n)}$$

- Equating the above with

$$\tilde{H}(\omega) = -j \sum_{n=0}^N h[n] e^{-j\omega(c+n)}$$

7.3 Types of Linear-Phase FIR Transfer Functions



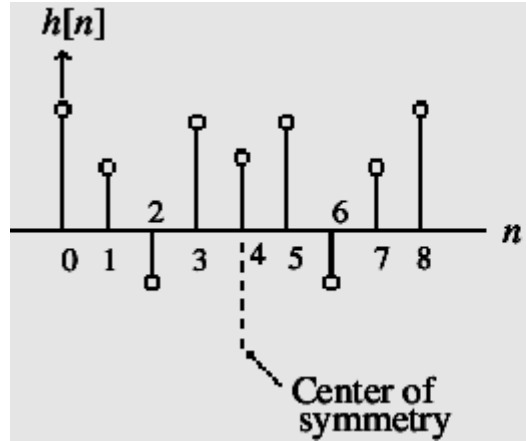
- We arrive at the condition for linear phase:
 $c = -N/2$ and $h[n] = -h[N-n]$, $0 \leq n \leq N$
- Therefore a FIR filter with an odd amplitude response will have linear-phase response if it has an **antisymmetric** impulse response.

7.3 Types of Linear-Phase FIR Transfer Functions

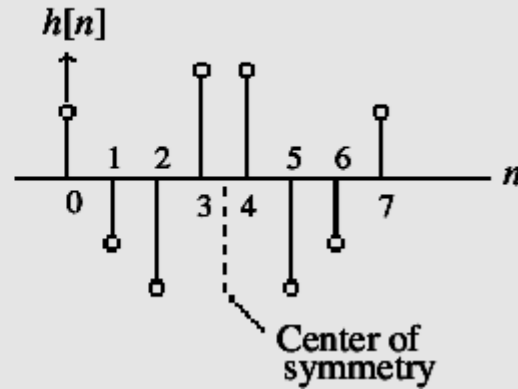


- Since the length of the impulse response can be either even or odd, we can define four types of linear-phase FIR transfer functions
- For an **antisymmetric** FIR filter of odd length, i.e., N even
$$h[N/2] = 0$$
- We examine next the each of the 4 cases

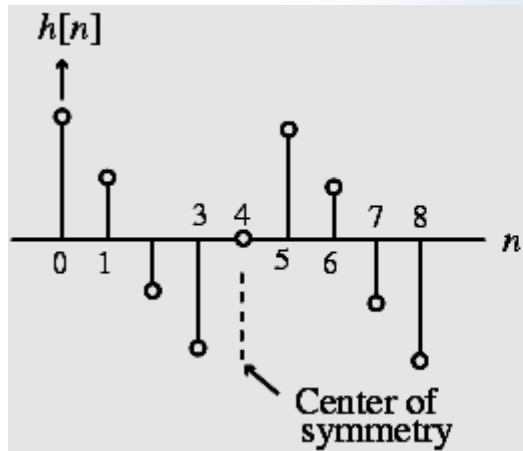
7.3 Types of Linear-Phase FIR Transfer Functions



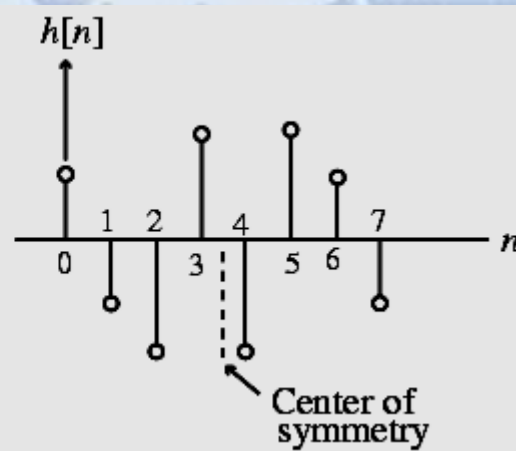
Type 1: $N = 8$



Type 2: $N = 7$



Type 3: $N = 8$



Type 4: $N = 7$

7.3 Types of Linear-Phase FIR Transfer Functions



Type 1: Symmetric Impulse Response with Odd Length

- In this case, the degree N is even
- Assume $N = 8$ for simplicity
- The transfer function $H(z)$ is given by

$$H(z) = h[0] + h[1]z^{-1} + h[2]z^{-2} + h[3]z^{-3} \\ + h[4]z^{-4} + h[5]z^{-5} + h[6]z^{-6} + h[7]z^{-7} + h[8]z^{-8}$$

7.3 Types of Linear-Phase FIR Transfer Functions



- Because of symmetry, we have $h[0]=h[8]$, $h[1] = h[7]$, $h[2] = h[6]$, and $h[3] = h[5]$
- Thus, we can write

$$\begin{aligned} H(z) &= h[0](1 + z^{-8}) + h[1](z^{-1} + z^{-7}) \\ &\quad + h[2](z^{-2} + z^{-6}) + h[3](z^{-3} + z^{-5}) + h[4]z^{-4} \\ &= z^{-4} \{ h[0](z^4 + z^{-4}) + h[1](z^3 + z^{-3}) \\ &\quad + h[2](z^2 + z^{-2}) + h[3](z + z^{-1}) + h[4] \} \end{aligned}$$

7.3 Types of Linear-Phase FIR Transfer Functions



- The corresponding frequency response is then given by

$$H(e^{j\omega}) = e^{-j4\omega} \{ 2h[0]\cos(4\omega) + 2h[1]\cos(3\omega) + 2h[2]\cos(2\omega) + 2h[3]\cos(\omega) + h[4] \}$$

- The quantity inside the braces is a real function of ω , and can assume positive or negative values in the range $0 \leq |\omega| \leq \pi$.

7.3 Types of Linear-Phase FIR Transfer Functions



- The phase function here is given by

$$\theta(\omega) = -4\omega + \beta$$

where β is either 0 or π , and hence, it is a linear function of ω in the generalized sense

- The group delay is given by

$$\tau(\omega) = -\frac{d\theta(\omega)}{d\omega} = 4$$

which is a constant group delay of 4 samples

7.3 Types of Linear-Phase FIR Transfer Functions



- In the general case for Type 1 FIR filters, the frequency response is of the form

$$H(e^{j\omega}) = e^{-jN\omega/2} \tilde{H}(\omega)$$

where the amplitude response $\tilde{H}(\omega)$, also called the **zero-phase response**, is of the form

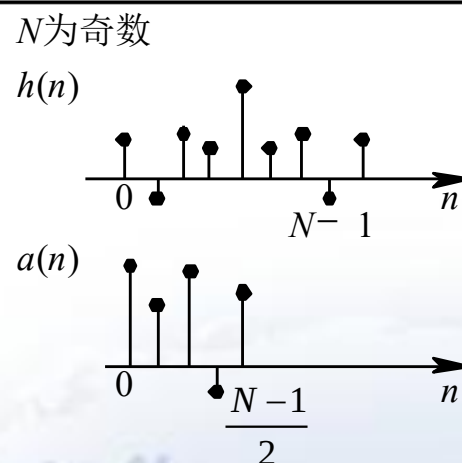
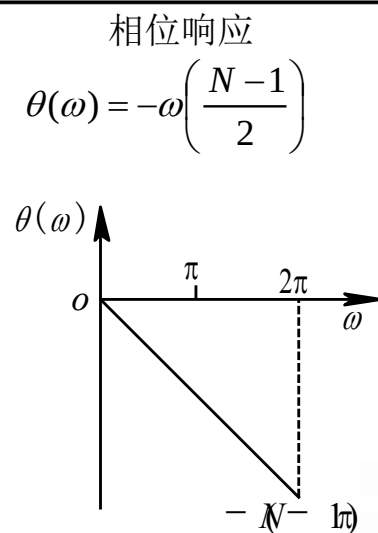
$$H(\omega) = h\left[\frac{N}{2}\right] + 2 \sum_{n=1}^{N/2} h\left[\frac{N}{2} - n\right] \cos(\omega n)$$

- See Section 7.3 for the forms of the other 3 types of transfer functions.

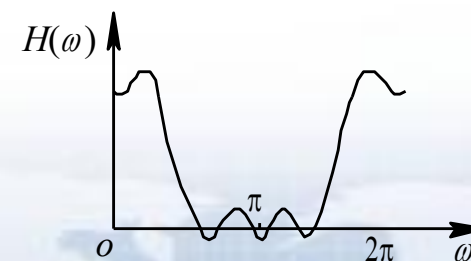
偶对称单位冲激响应

$$h(n) = h(N-1-n)$$

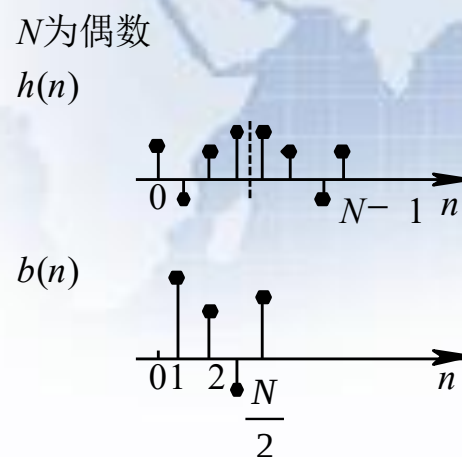
情况 1



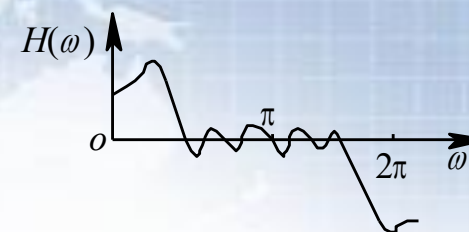
$$H(\omega) = \sum_{n=0}^{(N-1)/2} a(n) \cos n\omega$$



情况 2



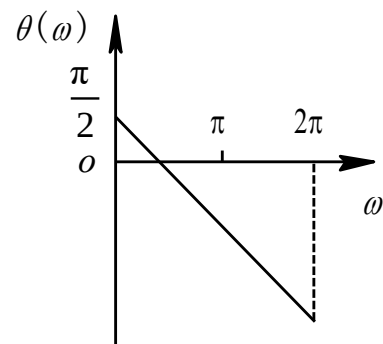
$$H(\omega) = \sum_{n=1}^{N/2} b(n) \cos \left[\left(n - \frac{1}{2} \right) \omega \right]$$



情况 3

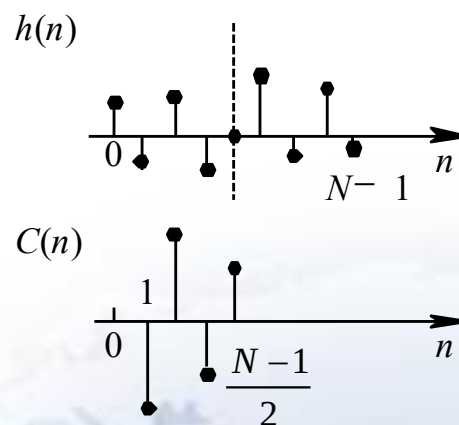
相位响应

$$\theta(\omega) = -\omega \left(\frac{N-1}{2} \right) + \frac{\pi}{2}$$

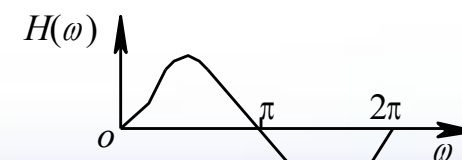


$$-\left(N - \frac{3}{2}\right)\pi$$

N 为奇数

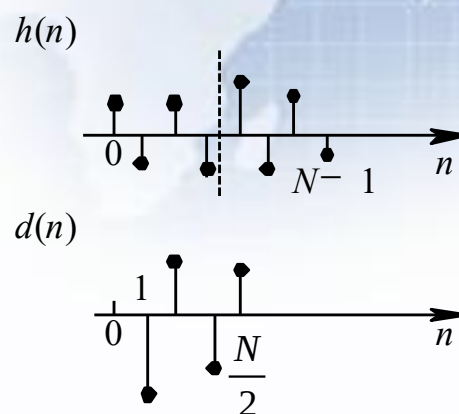


$$H(\omega) = \sum_{n=1}^{(N-1)/2} c(n) \sin(n\omega)$$

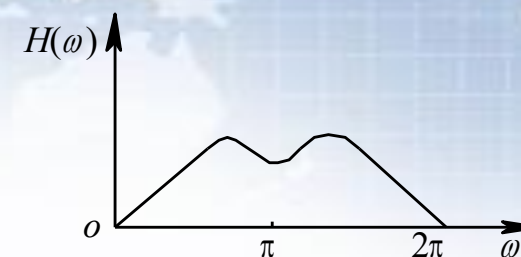


情况 4

N 为偶数



$$H(\omega) = \sum_{n=1}^{N/2} d(n) \sin \left[\omega \left(n - \frac{1}{2} \right) \right]$$



7.3 Types of Linear-Phase FIR Transfer Functions



Example - Consider

$$H_0(z) = \frac{1}{6} \left[\frac{1}{2} + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + \frac{1}{2} z^{-6} \right]$$

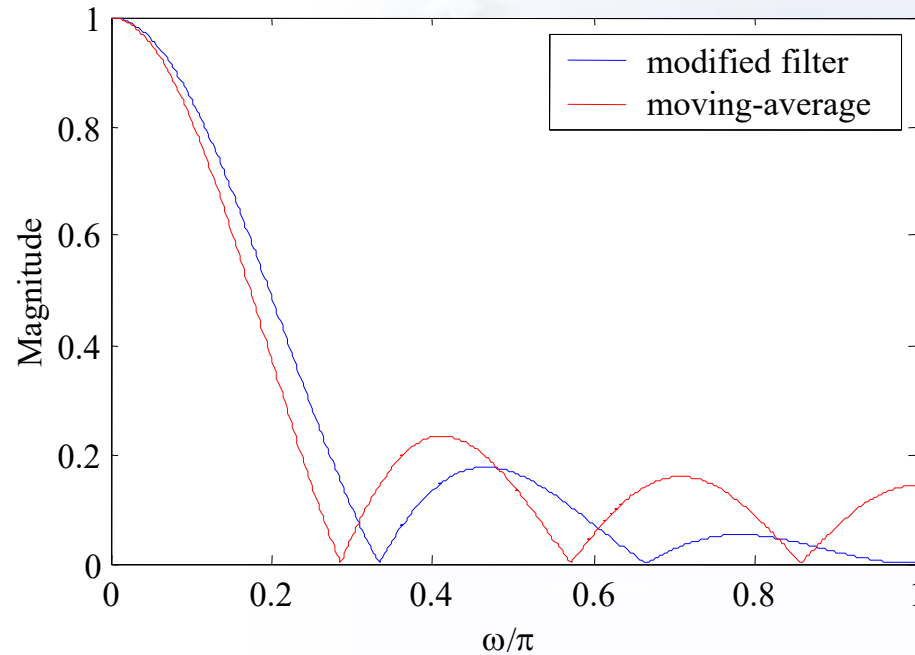
which is seen to be a slightly modified version of a length-7 moving-average FIR filter

- The above transfer function has a symmetric impulse response and therefore a linear phase response
 - Type 1

7.3 Types of Linear-Phase FIR Transfer Functions



- A plot of the magnitude responses of $H_0(z)$ along with the 7-point moving-average filter is shown below



7.3 Types of Linear-Phase FIR Transfer Functions



- Note the improved magnitude response obtained by simply changing the first and the last impulse response coefficients of a moving-average (MA) filter
- It can be shown that we can express

$$H_0(z) = \frac{1}{2}(1 + z^{-1}) \cdot \frac{1}{6}(1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5})$$

which is seen to be a cascade of a 2-point MA filter with a 6-point MA filter

- Thus, $H_0(z)$ has a double zero at $z=-1$, i.e., $(\omega = \pi)$

7.3.1 Zero Locations of Linear-Phase FIR Transfer Functions



- The zeros of the real-coefficient $h[n]$ is in mirror-image pairs.
 - For types 1 & 2 (symmetry)

$$H(z) = \sum_{n=0}^N h[n]z^{-n} = \sum_{n=0}^N h[N-n]z^{-n} \stackrel{\text{Let } m=N-n}{=} \sum_{m=0}^N h[m]z^{-N+m} = z^{-N} \sum_{m=0}^N h[m]z^m = z^{-N} H(z^{-1})$$
$$H(z) = z^{-N} H(z^{-1})$$

For types 3 & 4 (antisymmetry)

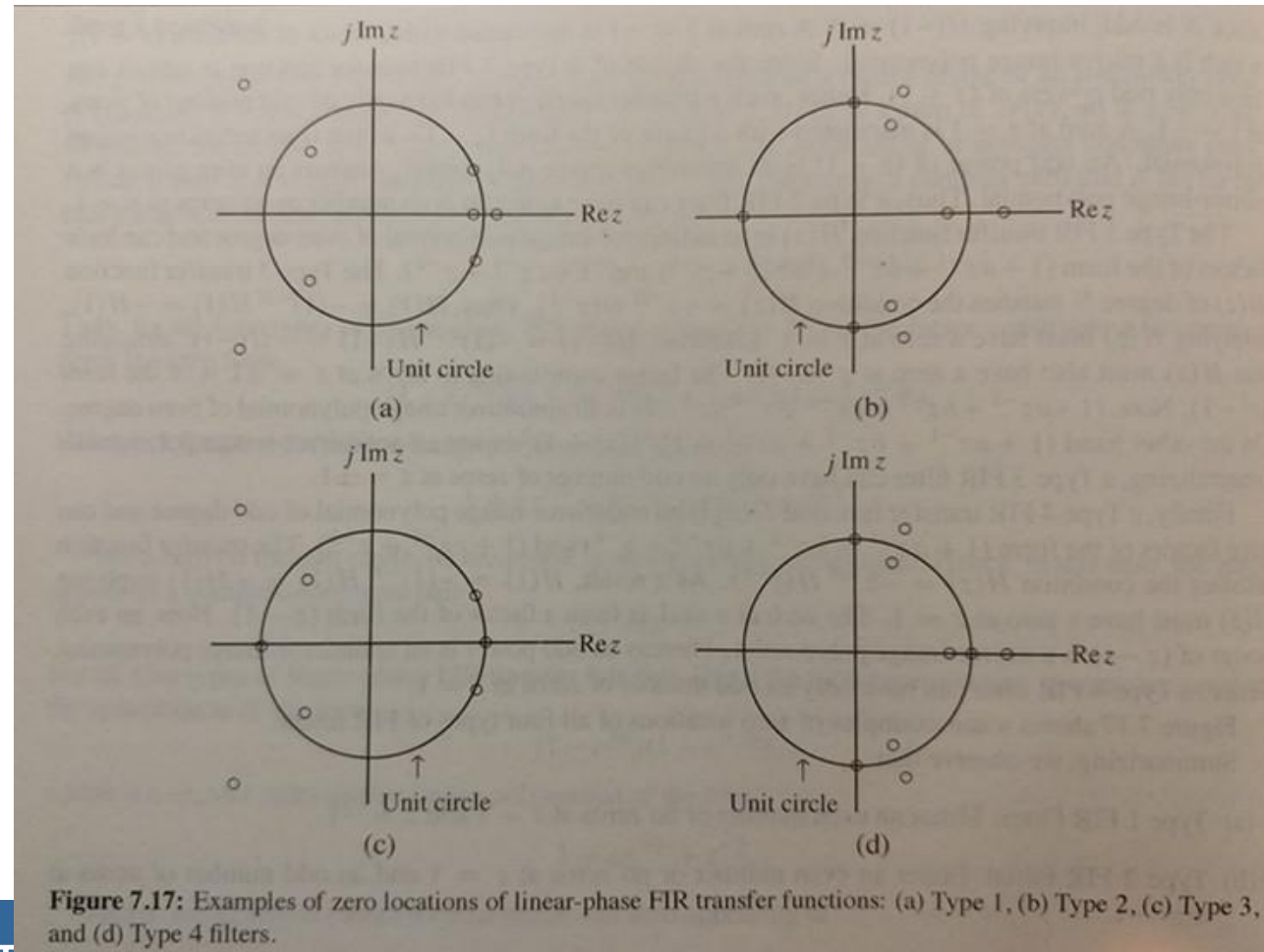
$$H(z) = -z^{-N} H(z^{-1})$$

If $H(z_0)=0$, then $H(1/z_0)=0$, i.e., zeros are in mirror image

7.3.1 Zero Locations of Linear-Phase FIR Transfer Functions



- Moreover, for **real** impulse $h[n]$, the zeros occur in **complex conjugate pairs**.



7.3.1 Zero Locations of Linear-Phase FIR Transfer Functions



- (1) Type 1 : Either an even number or no zeros at $z=1$ & $z=-1$.
- (2) Type 2 : Either an even number or no zeros at $z=1$, and an odd number of zeros at $z=-1$.
- (3) Type 3 : An odd number of zeros at $z=1$ & $z=-1$.
- (4) Type 4 : An odd number of zeros at $z=1$, and either an even number or no zeros at $z=-1$.

7.4 Simple Digital Filters



- Simple FIR Digital Filters

Lowpass; Highpass

- Simple IIR Digital Filters

Lowpass; Highpass; Bandpass; Bandstop

- Comb Filters

FIR Comb Filters; IIR Comb Filters

Homework



- Read the textbook from p.351 to 378
- Problems
7.4, 7.5, 7.10, 7. 21, 7.28, 7.41,7.43,7.60, 7.69

M7.3, M7.5, M7.6



THANKS